

MedAssist AI: Disease Prediction & Risk Assessment Model

Project Milestone 1 Report

Prepared by: Vinayak A S

Contents

1	Project Description	2
2	Dataset Used	2
3	Environment Setup	2
4	Data Exploration	2
5	Data Preprocessing	3
6	Proposed System	3
7	Training Pipeline & Evaluation	3
7.1	Training & Early Stopping	3
7.2	Results	4
7.3	Model Interpretability	5
8	Conclusion & Deployment	5

1 Project Description

MedAssist AI is a machine learning pipeline designed to predict medical conditions based on reported patient symptoms. Acting as an advanced differential diagnosis tool, the system analyzes combinations of symptoms and outputs the top candidate diseases with confidence probabilities. Unlike standard binary classification models, this project tackles an extreme multi-class classification problem (predicting 750+ diseases) with severe class imbalance. The core engine is built on XGBoost, utilizing gradient-boosted decision trees to capture complex, non-linear relationships between symptoms while penalizing overconfidence on rare diseases.

2 Dataset Used

About Dataset Description: The dataset contains disease names along with the symptoms faced by the respective patient. There are a total of 773 unique diseases and 377 symptoms, with ~246,000 rows. The dataset was artificially generated, preserving Symptom Severity and Disease Occurrence Possibility.

- **Original Scope:** 246,945 patient records.
- **Target Classes:** 773 initial unique diseases.
- **Features:** 377 initial unique symptoms.
- **Class Imbalance:** Highly imbalanced. A larger number of rows for a particular disease corresponds to its higher probability of occurrence in the real world. Common diseases (like the Flu) have thousands of records, while rare conditions may only have a single record.

Interpretation Info: Several distinct groups of symptoms might all be indicators of the same disease. There may even be one single symptom contributing to a disease in a row or sample. This is an indicator of a very high correlation between the symptom and that particular disease. Similarly, in a row, if the feature vector has the occurrence of a single symptom, it implies that this symptom has more correlation to classify the disease than any one symptom of a feature vector with multiple symptoms in another sample.

3 Environment Setup

- **Language & Frameworks:** Python, Scikit-Learn, Pandas, NumPy.
- **Core ML Library:** XGBoost (xgboost).
- **Visualization:** Matplotlib, Seaborn.
- **Hardware:** Trained on Google Colab using GPU (CUDA) acceleration to minimize computation time on high-dimensional data.

4 Data Exploration

Exploratory Data Analysis (EDA) was conducted to visualize the statistical distribution of the medical data.

- **Disease Frequency:** A bar chart of the Top 20 Most Frequent Diseases confirmed a heavy Pareto distribution, requiring specialized handling of “base rates.”
- **Symptom Frequency:** The EDA revealed that vague symptoms appear most frequently, meaning the model relies on combinatorial logic.

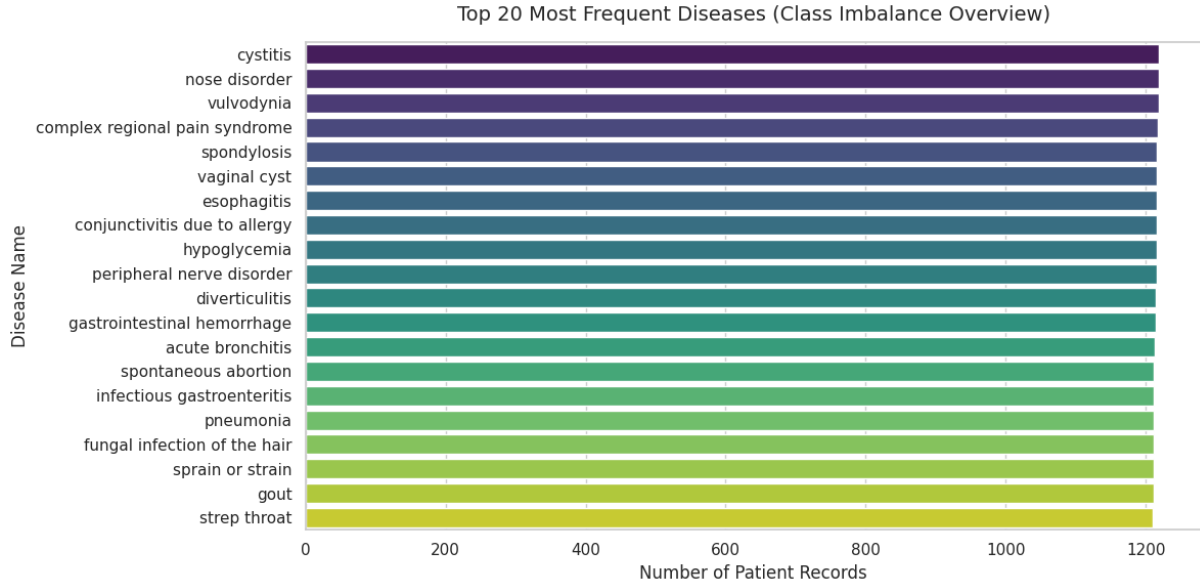


Figure 1: Top 20 Most Frequent Diseases

5 Data Preprocessing

A rigorous cleaning pipeline was implemented:

1. **Null Value & Integrity Check:** Scanned all 246,000+ rows to ensure structural completeness.
2. **Memory Optimization:** Downcast symptoms from int64 to uint8.
3. **Zero-Variance Filtering:** Dropped 49 non-varying symptoms.
4. **Rare Class Handling:** Removed 19 diseases with fewer than 2 samples.
5. **Label Encoding:** Transformed disease names into numeric IDs.

Result: 246,926 Rows | 328 Features | 754 Classes.

6 Proposed System

The architecture uses XGBoost customized for multi-class probability (`multi:softprob`).

- **Hardware Optimization:** Used `tree_method="hist"` and `device="cuda"`.
- **L2 Regularization (`reg_lambda=1.0`):** Prevented overfitting on rare classes.
- **Stochastic Hiding:** Used `subsample=0.8` and `colsample_bytree=0.7` to improve generalization.

7 Training Pipeline & Evaluation

The dataset was split using a stratified 80/20 train/test split. Stratification ensured that the rare disease distribution was perfectly mirrored in the test set.

7.1 Training & Early Stopping

The model halted at iteration 72 based on early stopping, avoiding overfitting.

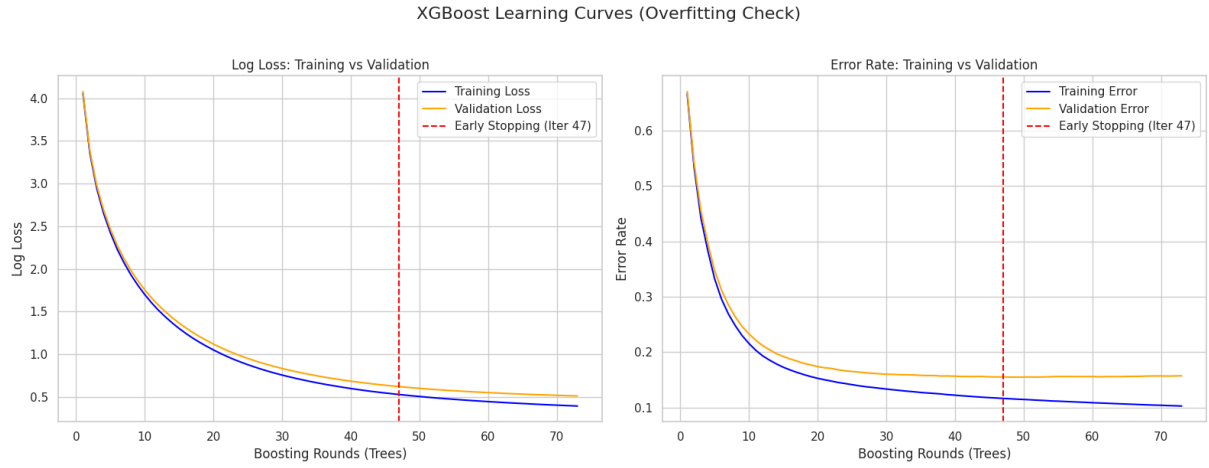


Figure 2: Training vs Validation Learning Curves

7.2 Results

- **Top-1/3/5 Accuracy:** 84.49% / 94.60% / 96.77%
- **Weighted F1-Score:** 0.8414
- **Macro F1-Score:** 0.6472
- **ROC-AUC:** 0.9994 (Weighted) / 0.9935 (Macro)

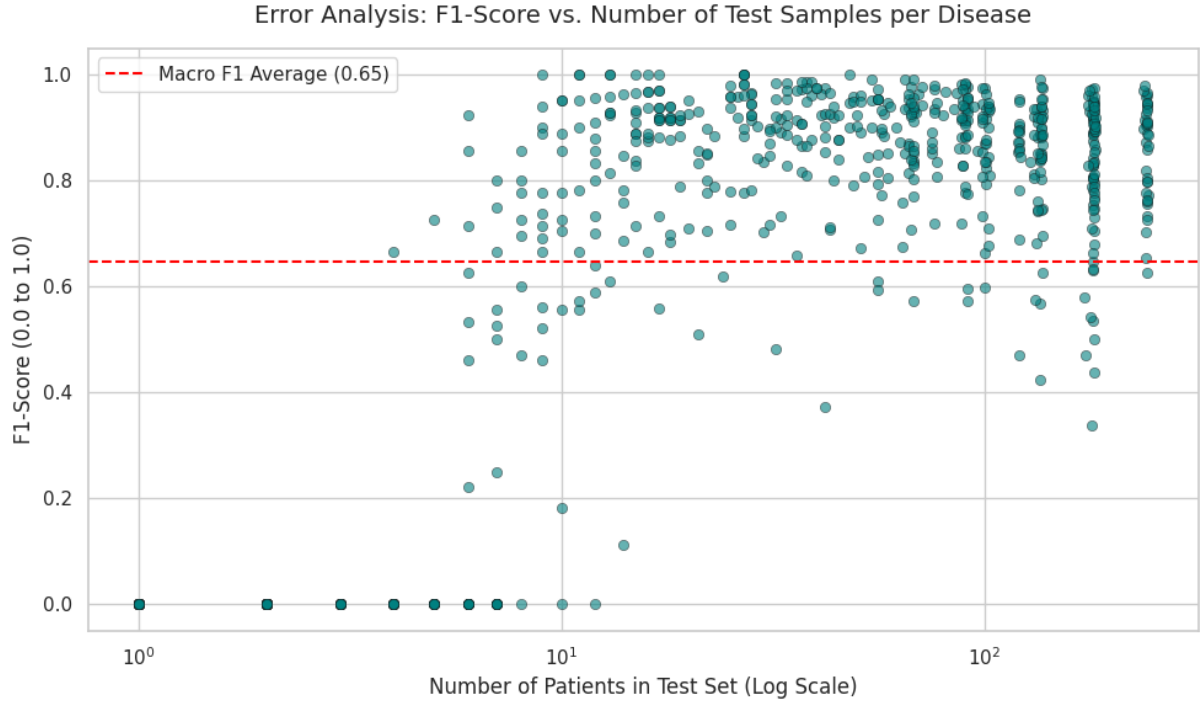


Figure 3: F1 Score vs Number of Patients

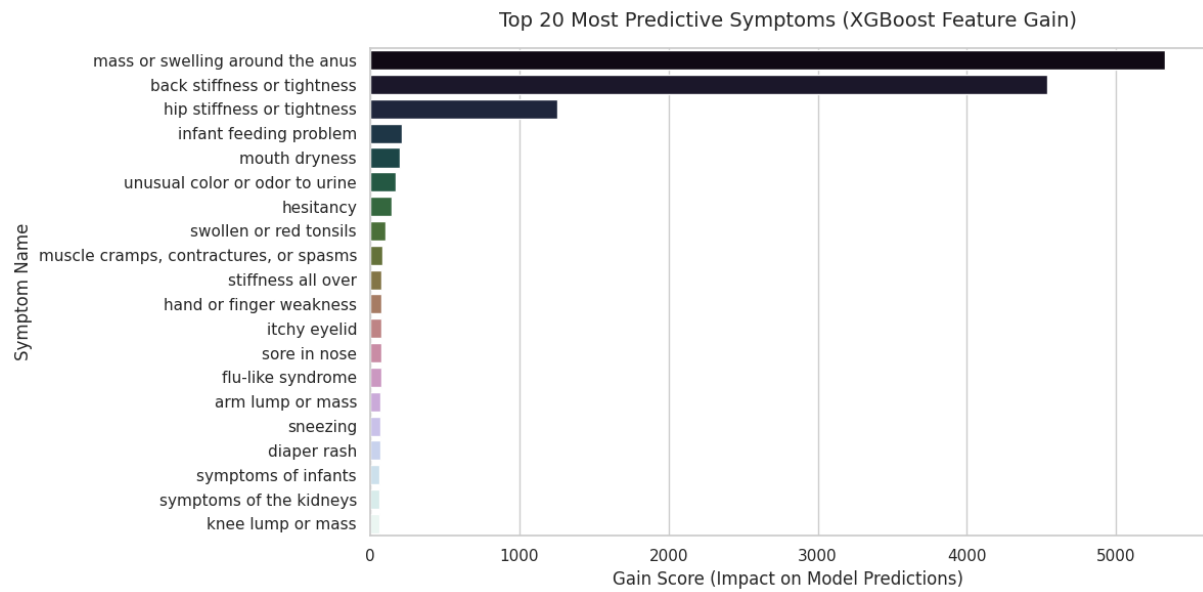


Figure 4: Top 20 Most Predictive Symptoms (Feature Gain)

7.3 Model Interpretability

8 Conclusion & Deployment

The model excels as a differential diagnosis tool, placing the correct disease in its Top 3 predictions 94.6% of the time. **Artifacts:** `disease_prediction_xgb.json`, `label_encoder.pkl`, `feature_columns.pkl`.